Rajarata University of Sri Lanka, Faculty of Applied Sciences

Department of Physical Sciences

MAP 2301 – Algebra

Tutorial 03

Answers

Answer all Submit **on or before 9th of June 2023**.

1. Let $\rho = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 3 & 5 & 4 & 6 & 9 & 7 & 1 & 8 & 10 & 2 \end{pmatrix}$ be a permutation in S_{10}.

   (i). Express ρ as a product of transpositions.
   (ii). What are the orders of ρ^{-1} and ρ^4?
   (iii). List the ρ – orbits and ρ^3 – orbits.

2. (a). Prove that the order of a permutation is the least common multiple of its disjoint cycles.
   (b). Let $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 6 & 5 & 4 & 1 & 2 & 7 \end{pmatrix}$ and $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 7 & 4 & 2 & 1 & 6 & 5 \end{pmatrix}$ be permutations in S_7.

   (i). Write α and β as products of transpositions.
   (ii). Find the order of $\alpha\beta$.
   (iii). Find a cycle σ in S_7 such that $\alpha^2\sigma = \beta^3$.

3. Let $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 6 & 5 & 4 & 1 & 2 & 7 \end{pmatrix}$ and $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 6 & 5 & 3 & 4 & 7 & 1 & 2 \end{pmatrix}$ be permutations in S_7.

   (i). What are the orders of α and β?
   (ii). Find α^{-1}, $\alpha\beta\alpha^{-1}$ and α^{40}.
   (iii). Find γ in S_7 such that $\gamma\alpha = \alpha\beta$.

4. Let $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 6 & 1 & 4 & 3 & 2 & 7 \end{pmatrix}$ and $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 6 & 4 & 3 & 5 & 7 & 2 & 1 \end{pmatrix}$ be permutations in S_7.

   (i). Find α^{-1} and $(\alpha\beta)^{50}$.
   (ii). Write $\alpha^2\beta^{-2}$ as a product of two disjoint 3-cycles.
   (iii). Find γ in S_7 such that $\gamma^2\alpha = \beta$.

5. Let $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 3 & 4 & 1 & 6 & 7 & 5 \end{pmatrix}$ and $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 6 & 5 & 1 & 3 & 4 & 2 \end{pmatrix}$ be permutations in S_7.

(i) Write α and β as products of disjoint cycles and calculate their orders.
(ii). Find αβ, β^{-1}, $\alpha\beta\alpha^{-1}\beta^{-1}$ and β^{25}.
(iii). Find non-trivial permutation ρ in S_7 such that $\alpha\beta\rho\beta^{-1}\alpha^{-1} = \rho$

6. Let $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 6 & 5 & 4 & 1 & 2 & 7 \end{pmatrix}$ and $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 6 & 5 & 3 & 7 & 2 & 1 & 4 \end{pmatrix}$ be permutations in S_7.

(i). Write α and β as products of disjoint cycles and calculate their orders.

(iii). Find non-trivial permutation ρ in S_7 such that $\alpha^{-2}\beta^{-2}\rho\beta^2\alpha^2 = \rho$.

7. (a). Prove that an even permutation in S_5 can be written as a product of 4-cycles.

(b). Let $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 1 & 5 & 4 & 7 & 8 & 6 \end{pmatrix}$ and $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 7 & 3 & 4 & 2 & 6 & 5 & 8 & 1 \end{pmatrix}$ be permutations in S_8.

(i). Find a permutation $\rho(\neq \omega, \alpha\beta, (\alpha\beta)^{-1})$ in S_8 such that $\alpha^{-1}\beta^{-1}\rho\beta\alpha = \rho$.
(ii). Find a cycle σ in S_8 such that $\sigma\alpha^4\sigma^{-1} = \beta^2$.

8. Let $\alpha = (1 \quad 2 \quad 4 \quad 3)(6 \quad 8 \quad 7)$ and $\beta = (1 \quad 3 \quad 4 \quad 2)\,(5 \quad 6 \quad 8 \quad 7)$ be permutations in S_8.

(i). Find α^{-1} and $(\alpha\beta)^{25}$.
(ii). What are the orders of α and β?
(iii). Find ρ in S_8 such that $\rho\beta\rho^{-1} = \beta^3$

① (i) $\rho = (1\ 3\ 4\ 6\ 7)(2\ 5\ 9\ 10)$

This can be written as a product of transpositions,

$$\rho = (1\ 3)(1\ 4)(1\ 6)(1\ 7)(2\ 5)(2\ 9)(2\ 10)$$

(ii) $\rho^{-1} = (7\ 6\ 4\ 3\ 1)(10\ 9\ 5\ 2)$

Order of $\rho^{-1} = 5 \times 4 = 20$

$$\rho^4 = [(1\ 3\ 4\ 6\ 7)(2\ 5\ 9\ 10)]^4$$

$$= (1\ 3\ 4\ 6\ 7)^4 \underbrace{(2\ 5\ 9\ 10)^4}_{=1}$$

$$= (1\ 3\ 4\ 6\ 7)(1\ 3\ 4\ 6\ 7)(1\ 3\ 4\ 6\ 7)(1\ 3\ 4\ 6\ 7)$$

$$\rho^4 = (1\ 7\ 6\ 4\ 3)$$

(iii) $\rho = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 3 & 5 & 4 & 6 & 9 & 7 & 1 & 8 & 10 & 2 \end{pmatrix}$

$1 \to 3 \to 4 \to 6 \to 7 \to 1$ orbit 1

$2 \to 5 \to 9 \to 10 \to 2$ orbit 2

$8 \to 8$ orbit 3

1st orbit $B_1 \Rightarrow (1\ 3\ 4\ 6\ 7)$

2nd orbit $B_2 \Rightarrow (2\ 5\ 9\ 10)$

3rd orbt $B_3 \Rightarrow (8)$

$\rho \in S_{10}$ has 3 orbits $= \{B_1, B_2, B_3\}$

$= (1\ 3\ 4\ 6\ 7)(2\ 5\ 9\ 10)(8)$

$$\rho^3 = [(1\ 3\ 4\ 6\ 7)(2\ 5\ 9\ 10)][(1\ 3\ 4\ 6\ 7)(2\ 5\ 9\ 10)]$$
$$[(1\ 3\ 4\ 6\ 7)(2\ 5\ 9\ 10)]$$

$$\rho^3 = (1\ 6\ 3\ 7\ 4)(2\ 10\ 9\ 5)(8)$$

Orbit 1 $= (1\ 6\ 3\ 7\ 4)$

orbit 2 $= (2\ 10\ 9\ 5)$

orbit 3 $= (8)$

(2) Order of a cycle =

(a) length of the cycle

Let α, β are two cycles of length m and n.

$\alpha^m = \varepsilon$, $\beta^n = \varepsilon$, ε-identity

Then k can be consider

as $k = l_1 m$

$$\alpha^{k} = \alpha^{l_1 m} = (\alpha^{m})^{l_1}$$

$$= (\varepsilon)^{l_1} = \varepsilon$$

Also $= l_2 n \Rightarrow \beta^{k} = \beta^{l_2 n}$

$$= (\beta^{n})^{l_2} = \varepsilon^{l_2} = \varepsilon$$

Since $\alpha\beta = \beta\alpha$

$$(\alpha\beta)^{k} = \alpha^{k}\beta^{k} = \varepsilon \cdot \varepsilon$$

$$= \varepsilon$$

$\alpha\beta$ become identity, its not necessary to be k is the order of $\alpha\beta$.

Let t is the order of α, β.

$$(\alpha\beta)^t = \varepsilon$$

$$\alpha^t \cdot \beta^t = \varepsilon$$

$$\alpha^t = \beta^{-t}$$

$\alpha = \beta$ or have common entities.

But α, β are disjoint

$\therefore \alpha, \beta$ are identities.

But m and n divides t.

$$t = k.$$

$\therefore$ Order of disjoint cycles is the least common multiplication of their length.

(b) (i)

$$\alpha = (1\ 3\ 5)(2\ 6) = (1\ 5)(1\ 3)(2\ 6)$$

$$\beta = (1\ 3\ 4\ 2\ 7\ 5) = (1\ 5)(1\ 7)(1\ 2)(1\ 4)(1\ 3)$$

(ii) $\alpha\beta = (1\ 3\ 5)(2\ 6)(1\ 3\ 4\ 2\ 7\ 5)$

$$= (1\ 5\ 3\ 4\ 6\ 2\ 7)$$

Order of $\alpha\beta = 7$

(iii) $\alpha^2 = (1 3 5)^2 (2\ 6)^2 = (1\ 3\ 5)(1\ 3\ 5)$

$$\alpha^2 = (1\ 5\ 3)$$

$$\beta^3 = (1\ 3\ 4\ 2\ 7\ 5)(1\ 3\ 4\ 2\ 7\ 5)(1\ 3\ 4\ 2\ 7\ 5)$$

$$\beta^3 = (1\ 2)(3\ 7)(4\ 5)$$

$$\alpha^2 \sigma = \beta^3$$

$\therefore \sigma = \alpha^{-2}\beta^{3}$

$= (3\ 5\ 1)(1\ 2)(3\ 7)(4\ 5)$

$\sigma = (1\ 2\ 3\ 7\ 5\ 4)$

3

(i) $\alpha = (1\ 3\ 5)(2\ 6)$

order $= 3 \times 2 = 6$

$\beta = (2\ 5\ 7)(1\ 6)$

order $= 3 \times 2 = 6$

(ii)

$\alpha = (1\ 3\ 5)(2\ 6)$

$\alpha\beta\alpha^{-1} = (1\ 3\ 5)(2\ 6)(2\ 5\ 7)(1\ 6)(5\ 3\ 1)(6\ 2)$

$= (1\ 7\ 6)(2\ 3)$

$$\alpha^{40} = \alpha^{6\times 6+4}$$

$$= \left(\left[(1\ 7\ 6)(2\ 3)\right]^6\right)^6 \left[(1\ 7\ 6)(2\ 3)\right]^4$$

$$= (1\ 7\ 6)^4(2\ 3)^4$$

$$= (1\ 7\ 6)^4\left((2\ 3)^2\right)^2$$

$$= (1\ 7\ 6)^4$$

$$= (1\ 7\ 6)^3(1\ 7\ 6)$$

$$= (1\ 7\ 6)$$

(iii) $\gamma\alpha = \alpha\beta$

$$\gamma = \alpha\beta\alpha^{-1}$$

$$= (1\ 7\ 6)(2\ 3)$$

(4) (i) $\alpha = (1\ 5\ 3)(2\ 6)$

$\therefore \alpha^{-1} = (3\ 5\ 1)(6\ 2)$

$\beta = (1\ 6\ 2\ 4\ 5\ 7)$

$\therefore \beta^{-1} = (7\ 5\ 4\ 2\ 6\ 1)$

$\alpha\beta = (1\ 5\ 3)(2\ 6)(1\ 6\ 2\ 4\ 5\ 7)$

$= (1\ 2\ 4\ 3)(5\ 7)$

$(\alpha\beta)^{50} = (\alpha\beta)^{8\times 6+2}$

$= \left[\left((1\ 2\ 4\ 3)(5\ 7)\right)^{8}\right]^{6}\left[(1\ 2\ 4\ 3)(5\ 7)\right]^{2}$

$= (1\ 2\ 4\ 3)^{2}(5\ 7)^{2}$

$= (1\ 2\ 4\ 3)(1\ 2\ 4\ 3)$

$= (1\ 4)(2\ 3)$

(ii) $\beta^{-2} = (7\ 5\ 4\ 2\ 6\ 1)(7\ 5\ 4\ 2\ 6\ 1)$

$= (1\ 5\ 2)(4\ 6\ 7)$

$\therefore \alpha^2 \beta^{-2} = [(1\ 5\ 3)(2\ 6)(1\ 5\ 3)(2\ 6)](1\ 5\ 2)(4\ 6\ 7)$

$$\underbrace{= [(1\ 3\ 5)]\overbrace{(1\ 5\ 2)(4\ 6\ 7)}^{\text{disjoint}}}_{\text{3 cycle}}$$

(iii)

$\gamma^2 \alpha = \beta$

$\gamma^2 \alpha = \beta$

$\gamma^2 = \beta \alpha^{-1}$

$= (3\ 5\ 1)(6\ 2)(1\ 6\ 2\ 4\ 5\ 7)$

$\gamma = (2\ 4\ 1)(7\ 3\ 5)$

(5)

(i) $\alpha = (1\ 2\ 3\ 4)(5\ 6\ 7)$

order $= 4 \times 3 = 12$

$\beta = (1\ 7\ 2\ 6\ 4)(5\ 3)$

order $= 5 \times 2 = 10$

(ii) $\beta^{25} = \beta^{(10\times 2)} \cdot \beta^{5}$

$= \{[(1\ 7\ 2\ 6\ 4)(5\ 3)]^{10}\}^{2}\ \beta^{5}$

$= \{(1)\}^{2} \cdot \beta^{5} = (1\ 7\ 2\ 6\ 4)^{5}(5\ 3)^{5}$

$= (5\ 3)^{5}$

$= \{(5\ 3)^{2}\}^{2} \cdot (5\ 3)$

$= \{(1)\}^{2} \cdot (5\ 3)$

$= (5\ 3)$

7 (a)

To prove that an even permutation in the symmetric group S_5 (the group of all permutations of five elements) can be written as a product of 4-cycles, we first need to understand the properties of even permutations and 4-cycles.

1. Even Permutation: A permutation is considered even if it can be expressed as a product of an even number of transpositions. A transposition is a permutation that swaps two elements and leaves the rest unchanged.

2. 4-cycle: A 4-cycle is a permutation that cyclically permutes four elements while leaving the rest unchanged. For example, (abcd) represents a 4-cycle where a maps to b, b maps to c, c maps to d, and d maps back to a.

Now, let's prove that any even permutation in S_5 can be written as a product of 4-cycles.

Proof:
Consider an arbitrary even permutation in S_5. Since it is an even permutation, it can be expressed as a product of an even number of transpositions. We'll aim to show that each transposition can be decomposed into a product of 4-cycles.

Let's consider a transposition (xy) where x and y are distinct elements in the set $\{1, 2, 3, 4, 5\}$. We want to decompose this transposition into a product of 4-cycles.

Case 1: x and y are not consecutive elements in the set $\{1, 2, 3, 4, 5\}$.
Without loss of generality, let's assume x < y. Then the 4-cycle (x, a, b, y) will swap x and y while leaving a and b unchanged, where a and b are any other two distinct elements in the set $\{1, 2, 3, 4, 5\}$. This 4-cycle can be expressed as:

(x, a, b, y) = (x, a)(a, b)(b, y)

Note that this 4-cycle will not affect any other element outside of $\{x, y, a, b\}$. Thus, we can decompose any transposition (xy) as a product of 4-cycles when x and y are not consecutive.

Case 2: x and y are consecutive elements in the set $\{1, 2, 3, 4, 5\}$.
Again, without loss of generality, let's assume x = 1 and y = 2. Then the 4-cycle (1, a, b, 2) will swap 1 and 2 while leaving a and b unchanged, where a and b are any two distinct elements in $\{3, 4, 5\}$. This 4-cycle can be expressed as:

(1, a, b, 2) = (1, a)(a, b)(b, 2)

As before, this 4-cycle will not affect any other element outside of $\{1, 2, a, b\}$. Therefore, any transposition (xy) where x and y are consecutive can also be decomposed into a product of 4-cycles.

Since every even permutation in S_5 can be expressed as a product of transpositions, and each transposition can be written as a product of 4-cycles (as shown in the two cases above), we have demonstrated that every even permutation in S_5 can be written as a product of 4-cycles.